Frequency-Adaptive Sharpness Regularization for Improving 3D Gaussian Splatting Generalization

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Abstract

Despite 3D Gaussian Splatting (3DGS) excelling in most configurations, it lacks generalization across novel viewpoints in a few-shot scenario because it overfits to the sparse observations. In this paper, we improve the generalization of 3DGS by reformulating its training objective with Frequency-Adaptive Sharpness Regularization (FASR). In contrast to previous methods employing foundation models or auxiliary supervision, we explicitly guide 3DGS to converge toward better generalization. Although Sharpness-Aware Minimization (SAM) similarly improves generalization of classification models by guiding optimization toward flat minima of the loss landscape, directly employing it to 3DGS is suboptimal due to the discrepancy in the tasks. I.e., it over-penalizes high-frequency regions while under-penalizing low-frequency ones. To address this, we estimate and penalize sharpness in a frequency-adaptive manner, which not only inherits the generalization ability of SAM to prevent floater artifacts, but also overcomes its drawbacks by preserving fine details that SAM tends to oversmooth. We show the versatility of our method through improvements over diverse baselines and datasets. Code will be released.

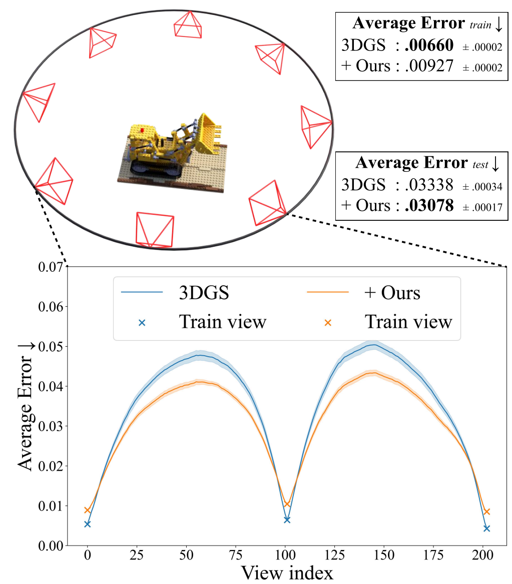


Figure 1. **Overview**. Our proposed optimization algorithm improves generalization. Given 8 training views rendered from the `lego` scene in Blender synthetic dataset [43], our method maintains low Average Error [46] across interpolated novel views, whereas the baseline exhibits overfitting. Plots are means and standard deviations over 10 runs.

1. Introduction

Reconstructing 3D scenes from multi-view 2D images has been a long-standing problem of interest. 3D Gaussian Splatting (3DGS) [23] achieves photo-realistic fidelity in novel view synthesis with real-time rendering. However, it requires densely captured input views, which are costly to obtain. In sparse-view settings, they often overfit to training views, resulting in poor generalization to novel views with unresolved details and floating artifacts. To improve generalization, previous approaches have adopted various strategies, such as integrating geometric priors from depth and flow estimators [8, 70], and leveraging dense correspondence prediction models [20]. Although these solutions show effectiveness, they rely on the assumption that the prior model performs well, which limits their general applicability. Moreover, research on fundamentally improving generalization has not been thoroughly investigated.

In this paper, we propose an optimization algorithm for 3DGS-based methods that greatly improves generalization (Fig. 1). The motivation is to optimize the model toward the flat minima in the loss landscape, ensuring better generalization. As illustrated in Fig. 2, sharp minima[1] exhibit a large gap between train and test loss, whereas flat minima

[1] We omit "local" for brevity.

show a much smaller gap, indicating better generalization. For example, artifacts like floaters are a result of the model overfitting to the training views. Consequently, they may be invisible from the training views but become immediately apparent with even slight perturbations on the model parameters[2]. This high sensitivity to perturbation implies a sharp minimum in the loss landscape, which has a large generalization gap.

Although Sharpness-Aware Minimization (SAM) theoretically and experimentally demonstrates that pursuing flatter minima leads to improved generalization in classification networks, *achieving flatter minima does not always guarantee better generalization in reconstruction tasks*. SAM applies a uniform sharpness penalty; therefore, applying it to 3DGS is suboptimal, because the importance of sharpness for accurate reconstruction varies across pixels. In particular, pixels with high-frequency details (e.g., edges) inherently induce a sharp loss landscape; even small changes in parameters cause a drastic change in the loss, thereby favoring sharp minima for accurate reconstruction. In contrast, pixels in low-frequency regions, the loss changes more gradually, favoring flat minima. To this end, we propose a frequency-adaptive sharpness regularization (FASR), an optimization algorithm that both estimates and penalizes sharpness in inverse proportion to frequency, enabling more appropriate estimation and balanced penalization. We show that the enhancement of our method is complementary to prior methods, thereby providing additional performance gains across them.

Our contributions are significant as follows:

- To the best of our knowledge, we present the first fundamental investigation into the relationship between loss landscape and generalization in the reconstruction task. We show that uniformly reducing sharpness oversmooths high-frequency details.
- Consequently, we propose an optimization algorithm for 3DGS by reformulating SAM in a frequency-adaptive manner, guiding the optimization toward better generalization.
- Thanks to the broad applicability of our method, we show generalization improvements on various baselines and datasets.

2. Related work

2.1. Sparse-view reconstruction

Reconstructing scenes from highly sparse inputs remains a fundamental challenge, as limited supervision often leads models to overfit training views, resulting in insufficient generalization in unseen views. Early efforts sought to extend neural radiance fields (NeRFs) by incorporating additional regularizations or auxiliary cues. For example, these studies [9, 46, 56, 64] introduce geometry- or depth-based constraints and frequency regularization to alleviate the underconstrained nature of sparse-view optimization. Although these approaches demonstrate partial effectiveness, they nonetheless inherit the slow training and rendering processes of NeRF.

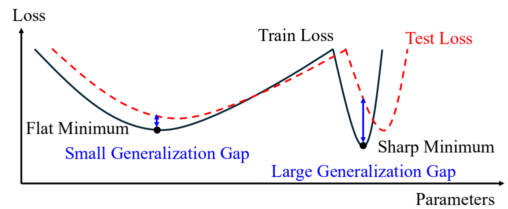


Figure 2. **Conceptual 1D Loss Landscape of Flat and Sharp Minima**[3]. Flat minimum better generalize then sharp minimum.

Recent studies have shifted to 3D Gaussian Splatting (3DGS), aiming for both real-time rendering efficiency and sparse-view fidelity. These studies [8, 20, 32, 62, 70, 72] leverage external priors such as depth [6, 48], correspondence [31], or flow [51], while another focuses on prior-free regularization of the representation itself [47, 67, 69]. Specifically, CoR-GS [67] optimizes multiple fields independently and reduces their disagreements, and DropGaussian [47] randomly drops Gaussians during training. Both methods are essentially stochastic ensemble-like regularizations; their improvement reduces when different subsets converge similarly. In contrast, our method encourages flat minima by adversarially finding the local maxima, leading to more effective and stable generalization.

Our approach is complementary: rather than altering the representation or adding priors, we fundamentally guide 3DGS to converge on a general solution in underconstrained sparse supervision.

2.2. Flatness and generalization

The relationship between the flatness of the loss landscape and model generalization has been extensively investigated in prior research. Keskar et al. [24] shows that converging to sharp minima leads to poor generalization, while Jiang et al. [22] identifies that sharpness is one of the most correlated indicators of generalization. Subsequent work [7, 18] demonstrated that averaging parameters along training trajectories can lead to flatter minima with better generalization.

Beyond averaging strategies, Sharpness-Aware Minimization (SAM) [12] explicitly estimates and reduces sharpness, inspiring subsequent work [2, 35, 36, 44, 53, 60] to analyze and improve upon its formulation. Moreover,

[2]Camera perturbations are a subset of Gaussian parameter perturbations, e.g., translating all Gaussians is equivalent to moving the camera.

[3]Figure is reproduced from Keskar et al. [24]. As empirically shown in Izmailov et al. [18], the test loss landscape tends to shift relative to the train loss landscape, while Liu et al. [40] showed that the loss landscape differs with the imbalance level of the dataset.

some works have analyzed the accuracy of estimated sharpness as a measure of generalization. They show that sharpness changes with parameter rescalings [10], and the appropriate sharpness estimation differs across different training setups and tasks [3], which hinders the correlation between sharpness and generalization. Consequently, recent work introduces normalization and invariant formulations to appropriately estimate sharpness, enabling a more reliable link between sharpness and generalization [16, 19, 27, 29, 55]. On the other hand, some works report counterexamples where sharper models generalize well, indicating that flatter minima are not always the optimal strategy for generalization [4, 10, 59].

Building on this perspective, we revisit the loss landscape sharpness in the context of the reconstruction task. We hypothesize that the optimal sharpness is not uniform but varies spatially, depending on the local frequency content of the signal. Therefore, instead of pursuing a universally flat minimum, we introduce a frequency-adaptive optimization strategy that appropriately estimates sharpness for the reconstruction task and allows sharp minima where they benefit high-frequency details.

2.3. Reconstructing with random perturbation

Random perturbation has been introduced into radiance field training for two primary purposes.

Unlike our approach, which aims to improve generalization, some studies employ perturbation for uncertainty quantification rather than relying on a deterministic approach. Stochastic or Bayesian formulations of NeRF have explicitly model radiance or density distributions [30, 50]. Subsequent work perturbs trained models to estimate epistemic uncertainty [15]. Similar ideas have been extended to 3DGS, where Gaussian parameters are sampled from learned distributions to render both images and calibrated uncertainty maps [1].

Another works exploit stochasticity to better optimization. These approaches inject noise into Gaussian parameters or query locations to escape poor local minima [13, 26] or enable a coarse-to-fine training effect [38]. Related approaches also perturb the sampling policy itself for acceleration, encouraging the model to focus on under-trained regions and thereby improving convergence [25].

While these stochastic approaches can improve convergence, they may also introduce unexpected artifacts (Appendix C provides more details). Therefore, we provide a more fundamental alternative by explicitly encouraging flat minima in an adversarial manner.

3. Method

We first provide preliminary on SAM [12] (Sec. 3.1). Then, we introduce an approach to directly apply SAM to 3DGS (Sec. 3.2). Finally, we present our proposed method in detail (Sec. 3.3). Fig. 3 illustrates an overview of our method.

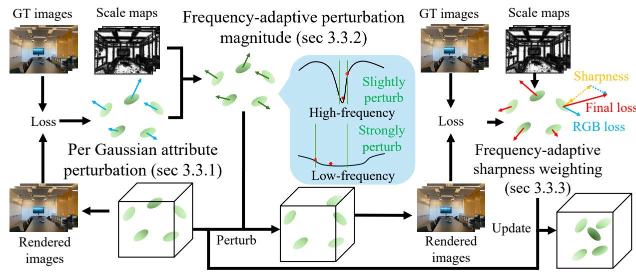


Figure 3. **Overview**. We first introduce per-Gaussian attribute perturbation (Sec. 3.3.1). We then propose frequency-adaptive perturbation magnitude and sharpness weighting that adjust their magnitude based on local frequency (Secs. 3.3.2 and 3.3.3).

3.1. Preliminary: Sharpness-Aware Minimization

Sharpness-Aware Minimization (SAM) [12] is an optimization method that improves generalization by explicitly encouraging solutions that lie in flat regions of the loss landscape. Along with minimizing the empirical loss $L(\boldsymbol{w})$ at a parameter point $\boldsymbol{w}$, SAM estimates the loss sharpness $[\max_{\|\boldsymbol{\epsilon}\|_2 \le \rho} L(\boldsymbol{w}+\boldsymbol{\epsilon}) - L(\boldsymbol{w})]$ within a neighborhood of radius ρ. The optimization then jointly minimizes both the empirical loss and this estimated sharpness, which is equivalent to optimizing the worst-case loss:

$$\begin{aligned} &\min_{\boldsymbol{w}} \left([\max_{\|\boldsymbol{\epsilon}\|_2 \le \rho} L(\boldsymbol{w}+\boldsymbol{\epsilon}) - L(\boldsymbol{w})] + L(\boldsymbol{w}) \right) \\ &= \min_{\boldsymbol{w}} \max_{\|\boldsymbol{\epsilon}\|_2 \le \rho} L(\boldsymbol{w}+\boldsymbol{\epsilon}) = \min_{\boldsymbol{w}} L^{\text{SAM}}. \end{aligned} \tag{1}$$

In practice, it is approximated via first-order Taylor expansion for the loss function around the current parameters, perturbing the parameters in the gradient direction with magnitude ρ:

$$\hat{\boldsymbol{\epsilon}}(\boldsymbol{w}) \triangleq \arg\max_{\|\boldsymbol{\epsilon}\|_2 \le \rho} L(\boldsymbol{w}+\boldsymbol{\epsilon}) \approx \rho \cdot \frac{\nabla_{\boldsymbol{w}} L(\boldsymbol{w})}{\|\nabla_{\boldsymbol{w}} L(\boldsymbol{w})\|_2}. \tag{2}$$

Then, the model parameters $\boldsymbol{w}$ are updated via Stochastic Gradient Descent (SGD) [45] step, with the gradient computed at the estimated local maximum $\boldsymbol{w} + \hat{\boldsymbol{\epsilon}}(\boldsymbol{w})$:

$$\boldsymbol{w} \leftarrow \boldsymbol{w} - \lambda_{\text{lr}} \cdot \nabla_{\boldsymbol{w}} L(\boldsymbol{w})|_{\boldsymbol{w}+\hat{\boldsymbol{\epsilon}}(\boldsymbol{w})},$$

where λ_{lr} is the learning rate at that step. Supporting the intuition of Fig. 2, it improves generalization by tightening the generalization bound based on sharpness derived from the PAC-Bayesian framework [11, 41, 49]:

Theorem 1. *For any* $\rho > 0$, *with training set* $\mathcal{S}$ *from data distribution* $\mathscr{D}$,

$$L_{\mathscr{D}}(\boldsymbol{w}) \le \max_{\|\boldsymbol{\epsilon}\|_2 \le \rho} L_{\mathcal{S}}(\boldsymbol{w}+\boldsymbol{\epsilon}) + h(\|\boldsymbol{w}\|_2^2/\rho^2),$$

where $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a strictly increasing function (under some technical conditions on $L_{\mathscr{D}}(\boldsymbol{w})$).

In practice, ρ is a hyperparameter.

The follow-up work WSAM [65] revisits sharpness as a regularization term by weighting the estimated sharpness:

$$\begin{aligned} L^{\text{WSAM}} &= \frac{1-2\gamma}{1-\gamma} L(\boldsymbol{w}) + \frac{\gamma}{1-\gamma} \max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} L(\boldsymbol{w}+\boldsymbol{\epsilon}) \\ &= L(\boldsymbol{w}) + \frac{\gamma}{1-\gamma} [\max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} L(\boldsymbol{w}+\boldsymbol{\epsilon}) - L(\boldsymbol{w})] \end{aligned} \tag{3}$$

When the weight hyperparameter γ is set to 0, the sharpness term vanishes and the optimization minimizes the empirical loss only. When $\gamma = 0.5$, WSAM is equivalent to SAM (Eq. (1)), while for $0.5 < \gamma < 1$, the sharpness term is emphasized. Accordingly, in the extended Theorem 1, $\max_{\|\boldsymbol{\epsilon}\|_2 \leq \rho} L_{\mathcal{S}}(\boldsymbol{w}+\boldsymbol{\epsilon}) = L_{\mathcal{S}}^{\text{SAM}}$ becomes $L_{\mathcal{S}}^{\text{WSAM}}$, and the function h is modified.

See Foret et al. [12] and Yue et al. [65] for the full theorem statement and proof.

3.2. Applying SAM to 3DGS

Let $\mathcal{G}_i$ be the i-th Gaussian, defined as $\mathcal{G}_i = (\boldsymbol{\mu}_i, \mathbf{q}_i, \mathbf{s}_i, \sigma_i, \mathbf{Y}_i^{\text{DC}}, \mathbf{Y}_i^{\text{AC}})$, where $\boldsymbol{\mu}, \mathbf{q}, \mathbf{s}, \sigma$, and $\mathbf{Y}$ represent the mean, rotation, scale, opacity and spherical harmonics (SH) coefficients, respectively, with $\mathbf{Y}_i^{\text{DC}}$ and $\mathbf{Y}_i^{\text{AC}}$ corresponding to the DC and AC terms of SH.

As in the 3DGS pipeline, we render 3D Gaussians $\mathcal{G} = (\mathcal{G}_i)_{i=1}^N$ from the camera view v and compute the loss L (Algorithm 1.2). The gradient of this loss guides the parameter to its local maximum (Algorithm 1.3-6). Similar to the SAM pipeline, we then evaluate the loss $\hat{L}$ at the local maximum and use this gradient to update the original parameters via the Adam optimizer [28] (Algorithm 1.7-8).

However, this direct application often leads to suboptimal performance, experimentally shown in Sec. 4.3.3. Unlike the classification task, in the reconstruction task, the curvature of the landscape is highly correlated with the image frequency.

Loss in high-frequency regions requires sharp minima for accurate reconstruction, making it sensitive to perturbations, whereas low-frequency regions favor flat minima and are less sensitive to them. Due to this differing sensitivity, a fixed $\rho_{\boldsymbol{\theta}}$ is problematic. In high-frequency regions, it leads to inaccurate sharpness estimation by causing large first-order approximation errors (Eq. 2). Conversely, in low-frequency regions, the perturbation is too weak to be meaningful, limiting the improvement of SAM. Moreover, SAM penalizes estimated sharpness with the fixed regularization weight $\gamma = 1$. It over-penalizes the high-frequency regions, which must remain sharp to preserve details. Concurrently, it under-penalizes the low-frequency regions, failing to sufficiently improve generalization.

Algorithm 1 Applying SAM to 3DGS

Input: Multi-view images $\tilde{I}^v$, where camera $v \in \mathcal{V}$
Output: Optimized $\mathcal{G} = (\boldsymbol{\mu}_i, \mathbf{q}_i, \mathbf{s}_i, \sigma_i, \mathbf{Y}_i^{\text{DC}}, \mathbf{Y}_i^{\text{AC}})$

1: **while** $\mathcal{G}$ not converged **do**
2: $\quad L \leftarrow \text{Loss}(\text{Render}(\mathcal{G}, v), \tilde{I}^v)$ *// Get loss*
3: $\quad$ **for all** $\boldsymbol{\theta} \in \mathcal{G}$ **do**
4: $\quad\quad \hat{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta} + \rho_{\boldsymbol{\theta}} \frac{\nabla_{\boldsymbol{\theta}} L}{\|\nabla_{\boldsymbol{\theta}} L\|_2}$ *// Ascent step*
5: $\quad$ **end for**
6: $\quad \hat{\mathcal{G}} \leftarrow (\hat{\boldsymbol{\theta}} \mid \boldsymbol{\theta} \in \mathcal{G})$ *// Local maximum*
7: $\quad \hat{L} \leftarrow \text{Loss}(\text{Render}(\hat{\mathcal{G}}, v), \tilde{I}^v)$ *// Get loss*
8: $\quad \mathcal{G} \leftarrow \mathcal{G} - \text{Adam}(\nabla_{\hat{\mathcal{G}}} \hat{L})$ *// Descent step*
9: **end while**

Based on this intuition, we hypothesize that the optimal perturbation magnitude $\rho_{\boldsymbol{\theta}}$ and regularization weight γ to effectively tighten the WSAM-extended version of the generalization bound in Theorem 1 vary in correlation with local frequency.

3.3. Frequency-Adaptive Sharpness Regularization

To address the limitation of Sec. 3.2, we introduce Frequency-Adaptive Sharpness Regularization (FASR), which adaptively modulates both the neighborhood radius $\rho_{\boldsymbol{\theta}}$ and the penalty weight γ inversely proportional to the image frequency.

LoG for frequency estimation. We calculate the local frequency at each pixel of the input image employing a multi-scale analysis inspired by blob detection using the Laplacian of Gaussian (LoG) [37]. For the grayscale image, we calculate the absolute LoG response at each candidate scale. We capture the optimal scale at each pixel by finding the first significant increase in its response curve that exceeds the threshold. The optimal scale is the smallest one that precedes such a rise; otherwise, we assign the maximum candidate scale. This process results in an optimal scale map $\boldsymbol{\Gamma}^v \in \mathbb{R}^{H \times W}$ at view v, with H and W denoting the height and width of the image, respectively. The local frequency is inversely proportional to this selected scale.

3.3.1. Per-Gaussian attribute perturbation

Unlike neural networks, 3DGS is an explicit model, which allows us to associate the local image frequency of each pixel with its corresponding Gaussian. To apply frequency adaptivity, we begin by perturbing the per-Gaussian attribute $\boldsymbol{\theta}_i \in \mathcal{G}_i$. Accordingly, Algorithm 1.4 becomes

$$\hat{\boldsymbol{\theta}}_i \leftarrow \boldsymbol{\theta}_i + \rho_{\boldsymbol{\theta}} \frac{\nabla_{\boldsymbol{\theta}_i} L}{\|\nabla_{\boldsymbol{\theta}_i} L\|_2}. \tag{4}$$

This step has the advantage of enabling more accurate sharpness estimation by normalizing the gradients of

Table 1. **Quantitative comparison on LLFF and MipNeRF-360 dataset**. Our method improves baselines across the board.

Method	LLFF (3 views)			MipNeRF-360 (12 views)		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
3DGS [ACM ToG'23]	19.802 ± .359	.6783 ± .0085	.2151 ± .0071	18.903 ± .179	.5499 ± .0036	.3734 ± .0042
+ Ours	**20.783** ± .300	**.7197** ± .0032	**.1965** ± .0034	**19.072** ± .176	**.5550** ± .0044	**.3609** ± .0039
CoR-GS [ECCV'24]	20.185 ± .142	.7015 ± .0040	.2029 ± .0035	19.515 ± .243	.5733 ± .0049	.3741 ± .0066
+ Ours	**20.862** ± .154	**.7283** ± .0042	**.1932** ± .0030	**19.754** ± .255	**.5845** ± .0054	**.3639** ± .0061
DropGaussian [CVPR'25]	20.461 ± .212	.7070 ± .0045	.2064 ± .0047	19.515 ± .202	.5722 ± .0042	.3657 ± .0036
+ Ours	**20.853** ± .227	**.7295** ± .0045	**.1969** ± .0040	**19.611** ± .191	**.5733** ± .0042	**.3605** ± .0042
NexusGS [CVPR'25]	21.048 ± .049	.7382 ± .0008	.1776 ± .0009	18.506 ± .098	.5222 ± .0031	.3587 ± .0021
+ Ours	**21.348** ± .078	**.7511** ± .0012	**.1714** ± .0011	**18.595** ± .117	**.5251** ± .0025	**.3579** ± .0023

the per-Gaussian attributes. Specifically, it mitigates the first-order approximation error from Gaussians in high-frequency regions, where the gradients tend to be large.

3.3.2. Frequency-adaptive perturbation magnitude

Going a step further, we adapt the perturbation magnitude according to the local image frequency. Specifically, for each Gaussian, we query the value of the optimal scale map at the projected center coordinate (x, y) on the 2D camera plane, $\gamma_i \leftarrow \boldsymbol{\Gamma}^v(x, y)$. Since this value is obtained in the 2D image space, we adjust it by multiplying the rendered depth at the same location, $d_i \leftarrow \boldsymbol{D}^v(x, y)$, $\boldsymbol{D}^v \in \mathbb{R}^{H\times W}$, and dividing by the focal length f, so that Gaussians located farther from the camera v are perturbed more strongly in 3D space. Finally, the resulting value is then multiplied by a predefined neighborhood radius to determine the final perturbation. Therefore, instead of Eq. (4) Algorithm 1.4 becomes

$$\hat{\boldsymbol{\theta}}_i \leftarrow \boldsymbol{\theta}_i + \gamma_i \cdot \frac{d_i}{f} \cdot \rho_{\boldsymbol{\theta}} \frac{\nabla_{\boldsymbol{\theta}_i} L}{\|\nabla_{\boldsymbol{\theta}_i} L\|_2}. \tag{5}$$

As a result, the perturbation magnitude is adaptive to the frequency of the corresponding Gaussian. This adaptation mitigates the first-order approximation error of Gaussians in high-frequency regions and perturbs strongly in low-frequency regions. Consequently, estimation of sharpness is more appropriate, thus improving generalization.

3.3.3. Frequency-adaptive sharpness weighting

Uniform penalty on sharpness across all attributes is suboptimal, as it tends to excessively penalize high-frequency regions that inherently correspond to sharp minima, losing their details. To address this issue, inspired by WSAM [65], we adapt the regularization weight according to the frequency of each Gaussian. From Eq. (3), the final gradient is a weighted combination of the gradient at the original point and the gradient at the perturbed point:

$$\nabla_{\hat{\boldsymbol{\theta}}_i} \hat{L} \leftarrow \frac{1-2\bar{\gamma}_i}{1-\bar{\gamma}_i} \nabla_{\boldsymbol{\theta}_i} L + \frac{\bar{\gamma}_i}{1-\bar{\gamma}_i} \nabla_{\hat{\boldsymbol{\theta}}_i} \hat{L}. \tag{6}$$

Here, $\bar{\gamma}_i \leftarrow 0.95 \cdot \gamma_i/\gamma_{\text{max}}$, where γ_{max} is the maximum candidate scale of the LoG kernel, and 0.95 is determined heuristically. We then update the parameter at the original point using this weighted gradient (Algorithm 1.8).

As a result, the sharpness penalty is reduced in high-frequency regions, preserving fine details, while applying a stronger penalty in low-frequency regions to facilitate better regularization.

4. Experiments

4.1. Experimental settings

Dataset. We evaluate our method on the LLFF [42] and MipNeRF-360 [5] datasets, following the experimental settings as previous works [47, 67, 70, 72], where the input resolution is 8× downsampled, and 3 and 12 input views are split for LLFF and MipNeRF-360, respectively. LLFF is a real-world dataset consisting of forward-facing scenes, while MipNeRF-360 is a large-scale dataset that captures 360-degree scenes.

Implementation. We selected the publicly available 3DGS [23] and its follow-up work as baselines. Specifically, among the follow-up works, we selected the state-of-the-art NexusGS [70], which leverages foundation models, and CoR-GS [67] and DropGaussian [47], which do not.

For all methods except 3DGS, we use the official repository. Since the official 3DGS does not target sparse-view reconstruction, we refactor it by referring to CoR-GS and DropGaussian. NexusGS does not provide optical flow for MipNeRF-360; we compute the flow using FlowFormer++ [51]. When applying regularization, we backpropagate the regularization term after computing the gradient of FASR.

Metrics. We use PSNR, SSIM [58], and LPIPS [68] as evaluation metrics, where LPIPS is computed with a

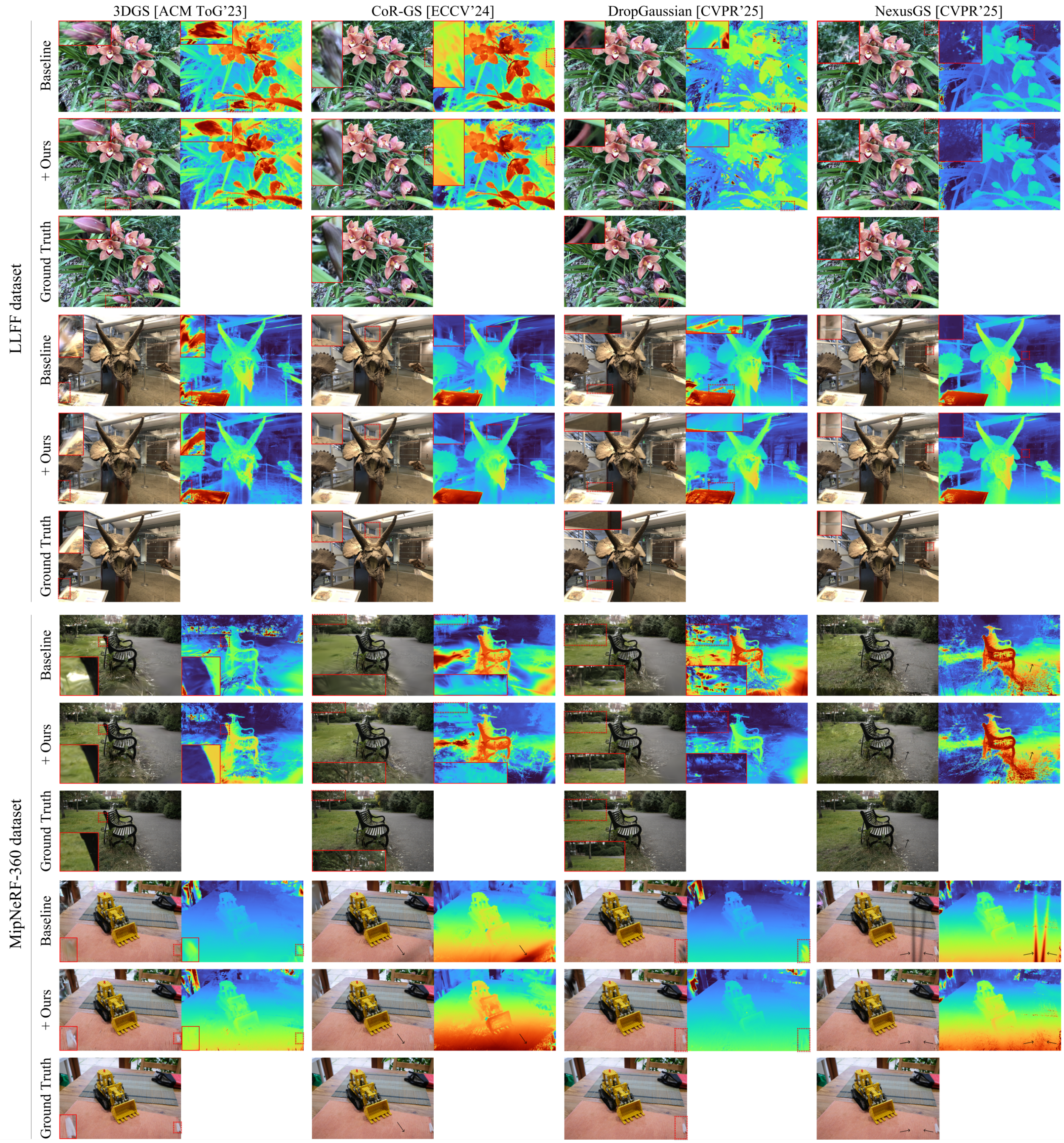
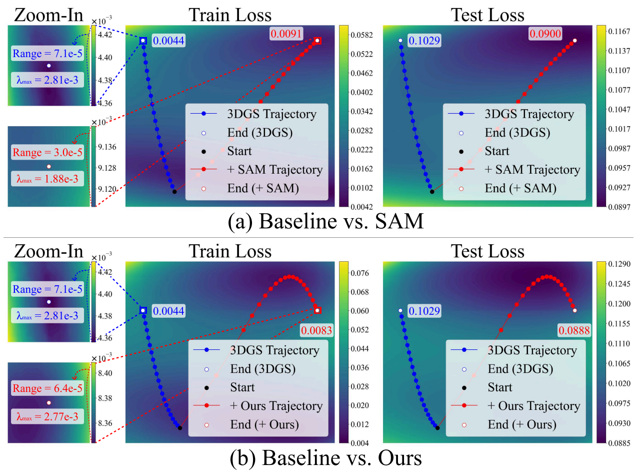


Figure 4. **Qualitative comparison on LLFF and MipNeRF-360 dataset**. Our method consistently improves reconstruction quality.

VGG network [52]. Additionally, we use Average Error (AVGE) [46], which is the geometric mean of PSNR, SSIM, and LPIPS. Considering the randomness of 3DGS, we conduce experiments with multiple runs.

4.2. Reconstruction quality

As shown in Tab. 1, our method achieves clear gains in PSNR, SSIM, and LPIPS across the LLFF and MipNeRF-360 datasets. When integrated into 3DGS and its variants, these improvements are achieved without any architectural changes or additional priors, enhancing generalization to unseen viewpoints. This reveals that our proposed optimization is a complementary enhancement that can be seamlessly integrated into current and future 3DGS-based frameworks. Fig. 4 show visual comparisons. Applying our method consistently improves the baselines by correcting



(a) Baseline vs. SAM

(b) Baseline vs. Ours

Figure 5. **Loss landscape visualization**. We compare the convergence of 3DGS and ours. Since the visualization yields a smoothed loss landscape, we also show a zoomed-in view near the convergence points of ours and the baseline. We measure global sharpness as the maximum eigenvalue $\lambda_{\max}$ of the Hessian.

Table 2. **Performance by covisibility level on LLFF dataset**. Improvement of our method is greater with higher view sparsity.

Covisibility level	AVGE ↓		
	3DGS	+ Ours	Δ
Covisibility 3	.0483 $\pm$.0093	.0417 $\pm$.0080	- .0066 $\pm$.0043
Covisibility 2	.0757 $\pm$.0156	.0644 $\pm$.0137	- .0114 $\pm$.0067
Covisibility 1	.0948 $\pm$.0256	.0806 $\pm$.0232	**- .0142** $\pm$.0074

geometric inaccuracies and reducing floating artifacts in novel views.

4.3. Analysis

4.3.1. Loss landscape visualization

To analyze the convergence behavior of our method, we visualize the reconstruction loss landscape and the corresponding optimization trajectories. We first train 3DGS for 5k iterations. Subsequently, we continue training for an additional 5k iterations with the densification process disabled, under three distinct settings: 3DGS optimization, SAM, and our proposed method. For visualization, we project the high-dimensional parameter trajectories onto a 2D plane using Principal Component Analysis with parameters from both trajectories.

SAM converges to flatter minima than 3DGS. In Fig. 5.a, the loss range between the local maximum and minimum is $2.37\times$, and the maximum Hessian eigenvalue is $1.49\times$ smaller. SAM also achieves a test loss of 0.0129 lower than 3DGS, narrowing the generalization gap from 0.0985 to 0.0809. However, our method converges to less flat minima than SAM. As shown in Fig. 5.b, the local loss range and maximum eigenvalue of ours are $1.11\times$ smaller and $1.01\times$ smaller than 3DGS, respectively. Interestingly, our method achieves test loss of 0.0141 lower than 3DGS and further narrows the generalization gap to 0.0805, outperforming SAM in generalization.

These results suggest that sharpness is not strictly correlated with generalization, supporting our hypothesis that sharpness of high-frequency pixel should be preserved. We further show that SAM tends to over-penalize high-frequency details, which leads to blurry results (Fig. 6).

Table 3. **Ablation study**. We report averaged results over multiple runs. Standard deviations are omitted due to space constraints. FAP denotes frequency adaptive perturbation magnitude, FAS denotes frequency adaptive sharpness weighting, and GAP denotes per-Gaussian attribute perturbation.

3DGS + Components				LLFF (3 Views)		
SAM	GAP	FAP	FAS	PSNR ↑	SSIM ↑	LPIPS ↓
✓	✓	✓	✓	**20.783**	**.7197**	**.1965**
✓	✓	✓	✗	20.560	.7116	.2023
✓	✓	✗	✓	20.570	.6980	.2142
✓	✓	✗	✗	20.390	.6977	.2174
✓	✗	✗	✗	20.198	.6958	.2095
✗	✗	✗	✗	19.802	.6783	.2151

4.3.2. Performance improvement by covisibility level

To examine where our method contributes to reconstruction quality, we measure AVGE across different covisibility levels. Following CoMapGS [20], we compute covisibility maps using MASt3R [31]. As shown in Tab. 2, the improvement of our method over the baseline 3DGS increases progressively as the covisibility decreases. This behavior aligns with our hypothesis that our method enhances generalization, particularly in under-constrained regions.

4.3.3. Ablation study

Directly applying SAM to 3DGS leads to degraded performance. This naive approach strongly perturbs Gaussians with large gradients, resulting in blurry reconstructions (Fig. 6, 6th column). Moreover, it increases the first-order approximation error, hindering the optimization process of SAM (Tab. 3, 7th row). Per-Gaussian perturbation, normalizes perturbation magnitude. Thus, it mitigates this issue, leading to improvements in some metrics (Tab. 3, 6th row), but the results remain blurry (Fig. 6, 5th column).

Frequency-adaptive sharpness weighting preserves sharpness in high-frequency details. Meanwhile, frequency-adaptive perturbation magnitude enables more faithful estimation by adaptively adjusting perturbation magnitude. Applying either component individually shows performance gains (Tab. 3, 4th and 5th rows). However, removing

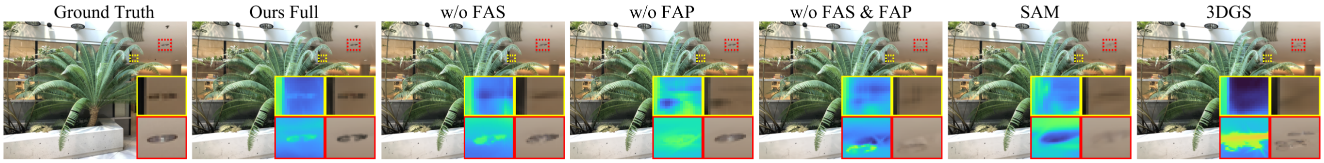


Figure 6. **Ablation study**. FAP denotes frequency-adaptive perturbation magnitude, FAS denotes frequency-adaptive sharpness weighting. "3DGS", "SAM", and "w/o FAS & FAP" produce inaccurate geometry (red box). All except "Ours Full" show blurry results (yellow box).

Table 4. **Applying FASR in late training phase**. For efficient training, we apply our method during the later iterations, denoted as Ours-L. Results are averaged over multiple runs using an RTX A5000. Standard deviations are omitted for brevity.

Method	LLFF (3 Views)			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time ↓
3DGS	19.802	.6783	.2151	**85** sec
3DGS + Ours	**20.783**	**.7197**	**.1965**	214 sec
3DGS + Ours-L	20.591	.7102	.2054	102 sec

either component degrades performance (Fig. 6, 4th and 3rd columns). Using both achieves a better generalization, balancing between sharpness reduction and detail preservation (Fig. 6, 2nd column and Tab. 3 3rd row).

4.4. Application

4.4.1. FASR in late training phase

The main limitation of SAM and its follow-up works is that they compute the loss gradient twice at each step, theoretically doubling the training time. Consequently, recent works [21, 39, 61] have focused on improving efficiency. Among them, Zhou et al. [71] demonstrated that SAM operates efficiently during the final few training epochs. Based on this finding, we apply our method during the last 12.5% of the total iterations, denoted as Ours-L. As shown in Tab. 4, Ours-L shows decreases of 0.192, 0.0095, and 0.0089 in PSNR, SSIM, and LPIPS, respectively compared to Ours, but still achieves improvements of 0.789, 0.0319, and 0.0097 over the baseline. Moreover, compared to the baseline, Ours increases the training time by $2.52\times$, Ours-L increases it by only $1.2\times$, making it more efficient.

The performance degradation of Ours-L occurs because 3DGS adaptively adjusts its number of learnable parameters through densification—a key difference from neural networks. This makes early training iterations influential, leading different results from Zhou et al. [71].

4.4.2. FASR to temporal sparsity

Additionally, we extend our method to dynamic scenes, the Neural 3D Video dataset [34], a fixed multi-view video dataset. Specifically, we conduct experiments with an online configuration [14, 17, 33, 63] where observation is temporally sparse, meaning that we can only access the current frame in the sequentially processed video frames. We applied our method to Yun et al. [66] with 3DGStream [54] backbone. When applying our method in the deformation process, we compute the FASR gradient on the deformed Gaussians and propagate it back to the deformation network. We update the residual map of Yun et al. [66] during the ascent step and the Gaussians during the descent step. Following their protocol, we select the first frame 3D Gaussians with the highest PSNR for initialization and compute the masked total variation (mTV) to measure temporal consistency, scaled by 100 for readability.

Table 5. **Applying FASR to online dynamic 3D Gaussians**. Our method is effective in dynamic scenarios under temporal sparsity, notably improving temporal consistency by reducing mTV.

Method	Neural 3D Video		
	PSNR ↑	SSIM ↑	mTV ↓
Yun et al. [66]	32.542	.9486	.1109
+ Ours	**32.622**	**.9497**	**.0989**

Our method improves temporal consistency and visual quality compared to the baseline (Tab. 5). This shows that our approach enhances generalization not only in the spatial domain but also in the temporal domain. Furthermore, Yun et al. [66] claims that one cause of temporal inconsistency is the inevitable noise in training datasets. Since SAM shows robustness on noisy training data, our finding aligns with this explanation.

5. Conclusion

In this work, we presented the first fundamental investigation into improving the generalization of 3D Gaussian Splatting by explicitly optimizing towards flat minima. We propose Frequency-Adaptive Sharpness Regularization (FASR), an optimization algorithm that reformulates Sharpness-Aware Minimization (SAM) in a frequency-adaptive manner. This reformulation overcomes the limitation of SAM in reconstruction tasks, achieving generalization with fine detail preservation. We hope this work inspires the research community to explore the link between sharpness and generalization in reconstruction fields.

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Frequency-Adaptive Sharpness Regularization for Improving 3D Gaussian Splatting Generalization

Supplementary Material

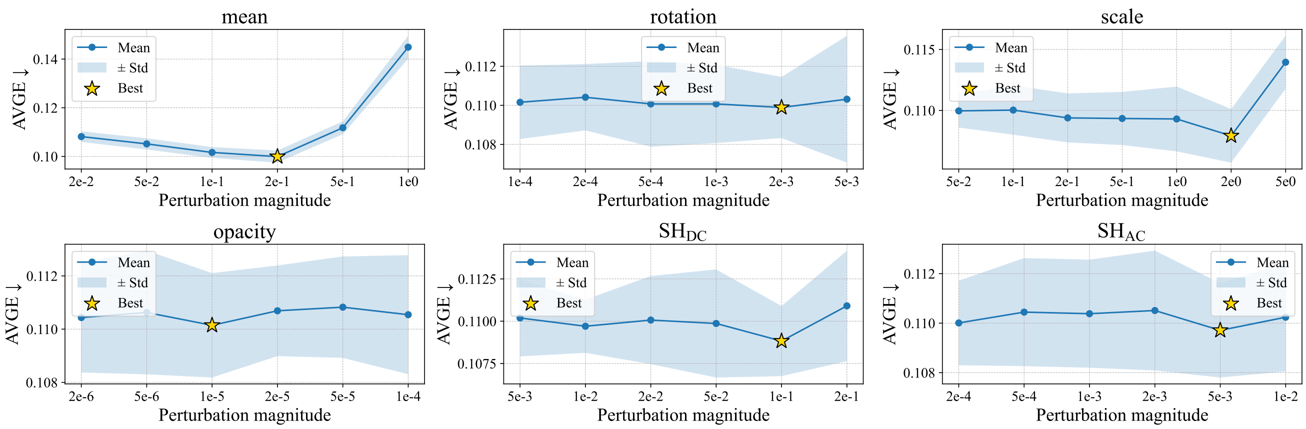

Figure 7. **Hyperparameter grid search, where each parameter is perturbed individually on the LLFF dataset**. Plots are means and standard deviations over 10 runs. We mark a star at the best hyperparameter value.

A. Hyperparameter search

In this section, we discuss selecting the hyperparameter neighborhood of radius ρ, i.e., the perturbation magnitude. Although our method scales ρ in a frequency-adaptive manner, ρ itself remains a core hyperparameter. We first search for the optimal perturbation magnitude by perturbing each parameter individually, as shown in Fig. 7.

However, the parameters are not independent and mutually influence each other. Consequently, the optimal ρ when perturbing all parameters simultaneously is typically smaller than the values found in this search. Therefore, we heuristically select smaller ρ values derived from this search as candidates and then fine-tune using this selection, perturbing all parameters simultaneously.

Selecting the optimal ρ remains a challenging problem in SAM-based methods. Recent studies, such as Wang et al. [57], explore learning the perturbation magnitude. Incorporating such approaches could be beneficial.

B. Perturbation per-arrtibutes

To analyze the contribution of each Gaussian attribute to the overall performance, we conducted an ablation study. We apply our method only to a single attribute or by excluding a single attribute. As shown in Tab. 6, we observe that while all attributes contribute positively to the performance, the Gaussian mean is the dominant contributor to the performance gain.

This finding suggests that instead of perturbing all attributes simultaneously—which complicates the tuning of ρ due to mutual parameter influence as shown in Appendix A—applying our method exclusively to the Gaussian mean could be a simplified approach, significantly reducing hyperparameter search complexity while likely retaining a substantial portion of the performance improvement.

Table 6. **Ablation study on the contribution of each Gaussian attributes**. The Gaussian mean is the dominant contributor to the performance improvement.

Method	LLFF (3 views)		
	PSNR ↑	SSIM ↑	LPIPS ↓
3DGS + Ours	**20.783** ± .300	**.7197** ± .0032	**.1965** ± .0034
w/o mean	20.053 ± .228	.6903 ± .0051	.2102 ± .0040
w/o rotation	20.656 ± .241	.7175 ± .0046	.1976 ± .0040
w/o scale	20.641 ± .235	.7171 ± .0049	.1987 ± .0035
w/o opacity	20.628 ± .218	.7175 ± .0044	.1977 ± .0033
w/o SH_{DC}	20.594 ± .242	.7152 ± .0052	.1972 ± .0036
w/o SH_{AC}	20.599 ± .205	.7169 ± .0041	.1973 ± .0034
w/ mean	20.561 ± .268	.7147 ± .0046	.1975 ± .0037
w/ rotation	19.854 ± .168	.6828 ± .0029	.2110 ± .0024
w/ scale	20.036 ± .222	.6871 ± .0048	.2093 ± .0037
w/ opacity	19.888 ± .188	.6803 ± .0045	.2124 ± .0033
w/ SH_{DC}	19.990 ± .202	.6867 ± .0048	.2115 ± .0034
w/ SH_{AC}	19.914 ± .186	.6827 ± .0042	.2114 ± .0033
3DGS	19.802 ± .359	.6783 ± .0085	.2151 ± .0071

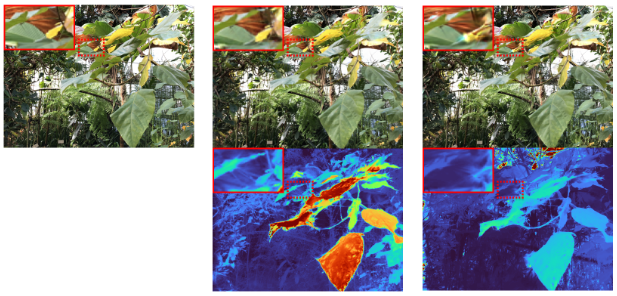

Figure 8. **Qualitative comparison of random perturbation and our method**. Random perturbation often introduce unexpected artifacts.

Method	LLFF (3 views) PSNR ↑	SSIM ↑	LPIPS ↓
3DGS + Ours	**20.783** ± .300	**.7197** ± .0032	**.1965** ± .0034
w/o AP	20.263 ± .289	.6990 ± .0031	.2056 ± .0027
w/ RP	20.174 ± .207	.6987 ± .0040	.2009 ± .0032
3DGS	19.802 ± .359	.6783 ± .0085	.2151 ± .0071

Table 7. **Quantitative comparison of random perturbation and adversarial perturbation**. AP denotes adversarial perturbation, and RP denotes random perturbation on Gaussian parameters.

C. Comparison to random perturbation

To validate our claims in Section Sec. 2.3 regarding the importance of adversarial perturbation, we compare our method with baselines that utilize random perturbation. For a fair comparison, we performed a grid search to find the optimal perturbation magnitude, same as Appendix A. As shown in Fig. 8, random perturbations often introduce unexpected artifacts.

Furthermore, we also compare a variant where the gradient is computed at a randomly perturbed position instead of the local maximum found adversarially. As shown in Tab. 7, all random perturbation methods improve 3DGS. However, their performance gains are significantly smaller than ours, indicating the importance of using adversarial perturbation.

D. More results

We report the quantitative results of each scene in Tab. 8. Our method method outperforms the baseline in most cases.

Table 8. **Per-scene quantitative results on LLFF and MipNeRF-360 dataset**. We report AVGE (lower is better) of each scene.

Method	LLFF (3 views)							
	fern	flower	fortress	horns	leaves	orchids	room	trex
3DGS	.0897 ±.00069	.1239 ±.00255	.0911 ±.00457	.1273 ±.00177	.1367 ±.00129	.1819 ±.00080	.0755 ±.00139	.0753 ±.00112
+ Ours	**.0765** ±.00062	**.1066** ±.00130	**.0780** ±.00412	**.1043** ±.00201	**.1170** ±.00092	**.1605** ±.00146	**.0686** ±.00154	**.0709** ±.00472
CoR-GS	.0806 ±.00075	.1088 ±.00275	.0754 ±.00120	.1205 ±.00093	.1427 ±.00212	.1735 ±.00138	.0723 ±.00089	.0698 ±.00283
+ Ours	**.0724** ±.00063	**.1014** ±.00198	**.0683** ±.00148	**.1075** ±.00154	**.1326** ±.00175	**.1603** ±.00090	**.0695** ±.00078	**.0614** ±.00225
DropGaussian	.0791 ±.00085	.1056 ±.00155	.0795 ±.00386	.1186 ±.00206	.1284 ±.00135	.1650 ±.00111	.0715 ±.00107	.0687 ±.00125
+ Ours	**.0717** ±.00071	**.1048** ±.00127	**.0783** ±.00293	**.1100** ±.00246	**.1176** ±.00097	**.1536** ±.00118	**.0700** ±.00143	**.0656** ±.00173
NexusGS	.0893 ±.00020	**.0981** ±.00023	.0498 ±.00030	.0909 ±.00043	.1072 ±.00025	.1383 ±.00044	.0755 ±.00052	.0810 ±.00051
+ Ours	**.0848** ±.00037	.0989 ±.00052	**.0475** ±.00081	**.0897** ±.00058	**.0992** ±.00039	**.1354** ±.00042	**.0696** ±.00039	**.0741** ±.00050

Method	MipNeRF-360 (12 views)						
	bicycle	bonsai	counter	garden	kitchen	room	stump
3DGS	.1763 ±.00318	.1408 ±.00300	.1464 ±.00218	.1302 ±.00185	**.1017** ±.00142	.1078 ±.00082	.2382 ±.00269
+ Ours	**.1736** ±.00158	**.1367** ±.00292	**.1400** ±.00158	**.1236** ±.00080	.1040 ±.00145	**.1045** ±.00164	**.2371** ±.00490
CoR-GS	.1732 ±.00291	.1263 ±.00204	.1310 ±.00111	.1297 ±.00317	.1053 ±.00240	**.0931** ±.00164	.2214 ±.00721
+ Ours	**.1697** ±.00245	**.1176** ±.00166	**.1241** ±.00124	**.1181** ±.00257	**.1048** ±.00232	.0990 ±.00306	**.2094** ±.00498
DropGaussian	.1646 ±.00117	.1301 ±.00182	.1356 ±.00121	.1218 ±.00108	.0978 ±.00209	.1025 ±.00375	.2212 ±.00318
+ Ours	**.1644** ±.00136	**.1295** ±.00211	**.1346** ±.00137	**.1193** ±.00155	**.0977** ±.00223	**.1000** ±.00207	**.2116** ±.00354
NexusGS	.1818 ±.00101	.1308 ±.00127	.1563 ±.00102	.1249 ±.00025	**.0921** ±.00045	**.1525** ±.00288	.2352 ±.00203
+ Ours	**.1782** ±.00268	**.1284** ±.00131	**.1508** ±.00097	**.1241** ±.00057	.0947 ±.00056	.1562 ±.00290	**.2336** ±.00103